

Five Steps to Becoming a
Cool Congregation

www.coolcongregations.org



Interfaith Power & Light

A Religious Response to Global Warming



Interfaith Power & Light **COOL CONGREGATIONS**

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Cool Congregations Challenge Flyer

Contest Kit Flyer

EPA Energy Star for Congregations

VERSION 1.1, Fall 2019

Comments or suggestions welcome at programs@interfaithpowerandlight.org

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Special thanks to Sarah Paulos of Iowa Interfaith Power & Light for her dedication in developing, testing, and teaching the Cool Congregations Stewardship Program. Information about the *Household Cool Congregations Contest* is included herein and was the inspiration for the development of the *Certified Cool Congregations* program and the *Cool Congregations Challenge*.

Introduction

to Cool Congregations



Climate change is one of the biggest threats facing humanity today. Every major religion has a mandate to care for Creation and to act as good stewards and preserve life for future generations. Many denominations also now have [climate statements, which you can find on our website](#). The Cool Congregations program is designed to support faith communities as they “walk the talk” by reducing their own carbon footprint, thus helping to cool the planet. A side benefit of the program is the ‘multiplier effect,’ as congregants are encouraged to model the same energy saving behavior at home that they see at their congregation.

The Environmental Protection Agency’s ENERGY STAR estimates that if America’s 370,000 congregations cut energy use just 20% it would save nearly \$630 million per year that could be applied to missions and other priorities, while providing more than 3.6 billion “new” kWh electricity per year without additional cost or pollution. These energy savings

would prevent more than 2.6 million tons of greenhouse gas emissions per year — the equivalent of eliminating emissions from 480,000 cars, or planting 60,000 trees every year!

IPL offers both **Certification** and a prize money **Challenge** to exemplary Cool Congregations. IPL's annual *Cool Congregation Challenge* is a fun and easy way for congregations to compete for recognition and prize money. Certification demonstrates your leadership in your community by showing it's possible to reduce emissions, care for our common home, and save money in the process. For more information contact programs@interfaithpowerandlight.org.

We're very pleased that you're interested in joining the ranks of Cool Congregations nationwide. Thank you for providing inspiration and leadership to your congregation. By empowering your members to act, you will also bring more voices to the growing interfaith movement that is demanding accountability and leadership from Washington, DC on behalf of the Earth and the common good.

Benefits of Becoming a Certified Cool Congregation

Becoming a Certified Cool Congregation is a wonderful way for faith communities to save money on energy bills, respond to climate change, and have more money to spend on their missions. It's a win-win all the way around. This guide outlines five steps to becoming a *Certified Cool Congregation* and provides details about how to participate in our annual *Cool Congregations Challenge* for prize money, as well. There is also information about the Household *Cool Congregations Contest*, which is a fun way to extend your impact by inspiring your members to save energy at home.



Five Steps to Becoming a Cool Congregation

A *Certified Cool Congregation* is a congregation or place of worship of any faith that cuts its carbon footprint by at least 10%. Four levels of recognition are available, for 10%, 20%, 30%, and 40% carbon footprint reduction. Congregations can achieve higher and higher levels of certification with each passing year.

There is a no fee to apply for certification, although we request that participating congregations donate to PL from your energy cost savings so PL can continue to help more congregations go green. A *Certified Cool Congregation* receives a paper certificate, a listing on the coolcongregations.org website, inclusion in the US Green Building Council's Green Building Information Gateway, and an option of purchasing an attractive eco-friendly engraved wooden wall plaque.

Energy savings can be earned over any period of time. Energy savings and carbon reductions that occur as a result of energy efficiency, conservation, or renewable energy projects can be added together to determine total carbon savings.

In This Guide

Step 1: Learn & Plan

Learn how to form a green team or join an existing one, think about your goals, reflect on your personal motivations, align with your denomination's climate statement, and write your green team mission statement.

Step 2: Find Ideas

Get ideas for what you might want to do by learning about cost effective strategies for using less energy and saving money on your energy bills. More than anything, Cool Congregations is a community. IPL is a national organization, with 40-state affiliates. Also, learn how to join the Cool Congregations mailing list and find frequently updated resources on the coolcongregations.org website. Partner resources are also in this section.

Step 3: Get Help

When you are part of Interfaith Power & Light's Cool Congregations program, there are always people to help. Use the resources in this section to find the help you need to plan and complete your project.

Step 4: Make Improvements

Learn about energy audits, where to get one, how to do one yourself if necessary, how to measure your “baseline” energy use using various online calculators, and how to find help to devise an action plan and budget. Also in this section, learn about creative ideas for funding your project and finding volunteers.

Step 5: Apply for Certification

Earn the recognition that your congregation and team deserves. This section walks you through the certification process and provides instructions for filling out the online application form.

For Fun: Take the Cool Congregations Challenge and Win Prizes

The *Cool Congregations Challenge* is a fun, annual juried contest **for congregational facilities** that awards cash prizes in five categories including: Cool Congregations Planner, Energy Saver, Renewable Role Model, Sacred Grounds Steward, Community Inspiration. Congregations can participate in both the certification process and the *Cool Congregations Challenge* using the same project if they choose. Find out how to enter inside this guide.

For even MORE fun: The Household Cool Congregations Contest

The combined carbon footprint of all your congregation's members' homes is probably larger than the carbon footprint of your congregational facility. Engage your members with a fun Cool Congregations Contest to see who can reduce their footprint the most! Give out prizes and celebrate! Add up the total reductions of your congregants and see if they can beat the worship facility in lowering their carbon footprint! A little friendly competition is exciting and fun! Order your Household Cool Congregations Contest kit at www.coolcongregations.org to get all you need to run a successful contest.

Handouts

As one of the leaders of a Cool Congregation, you'll have to become both an educator and an organizer of sorts. This section contains many useful and printable handouts designed to help bring people together, educate them, and rally them to your cause!

About Interfaith Power & Light

Cool Congregations is a program of Interfaith Power & Light (IPL). IPL inspires and mobilizes people of faith and conscience to take bold and just action on climate. For more than twenty years IPL has been doing this through its network of 40-state affiliates that reach an estimated 24,000 congregations serving 5 million people.

Cool Congregations walk the talk, with small- and large-scale projects that reduce dependence on fossil fuels and at the same time model energy efficiency through conservation, renewable energy, sustainable grounds, and water conservation. IPL encourages people of faith and conscience to stand up and speak on behalf of our children's future and our common home through state and federal civic engagement opportunities. Climate actions include such things as sermons about climate change; mass letter writing campaigns to policy makers; opinion pieces; lobby days; participation in EPA public commenting, and other ways to be heard and influence public policy.

Climate change is one of the biggest threats facing humanity today. The integrity of Creation and life as we know it is jeopardized by our continued dependency on fossil fuels for energy. We build on the common tenets of all faiths to engage congregations in the sacred work of preserving and protecting the Earth and future generations. Learn more about Interfaith Power & Light by visiting our website, www.interfaithpowerandlight.org.

Step 1

Learn

Learning About Climate Change

The path to becoming a Cool Congregation goes beyond swapping out light bulbs and insulation. The underlying motivation for Cool Congregation is a religious response to climate change by becoming stewards of God's Creation through personal action.

Reducing climate change pollution (carbon footprint) is a big part of program. So too is educating and inspiring members of your congregation and the community about the problem of climate change and the solutions.

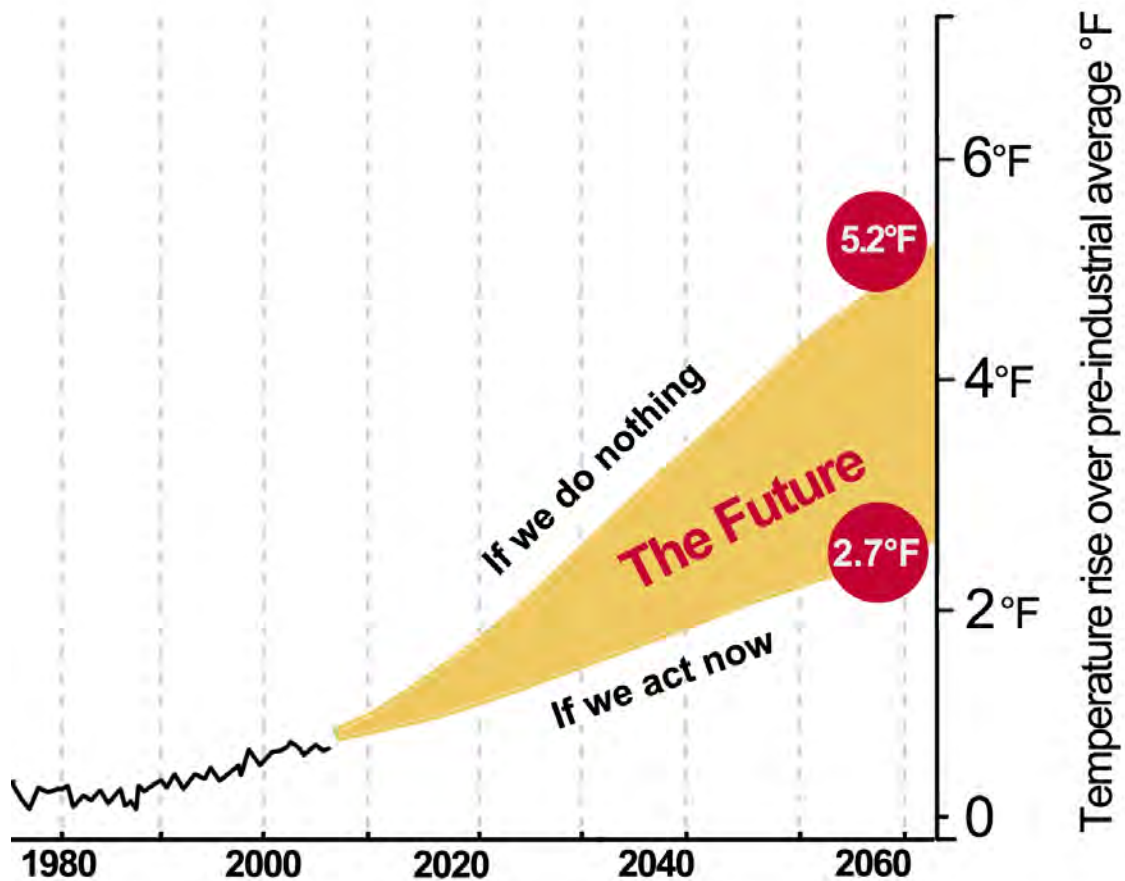
The impacts of climate change are happening faster than predicted. We can see the signs all around us rising sea levels washing away homes, extreme weather events such as hurricanes, droughts, floods, wildfires have become more dangerous and frequent and are threatening life as we know it in our own communities.

How much will the world will warm in the next 10, 20, 100, 200 years? If we do nothing, the predictions about climate change are severe. But if we all band together and reduce climate change pollution by transitioning to clean, renewable energy, the warming will be much less and the future will look much brighter for this and future generations.

All scientists agree that if we adjust our course now and move actively towards renewable energy and away from fossil fuels, the consequences of climate change will be more manageable. No matter what we do, we are now locked in to some extent of warming. The goal is to lessen the impact on future generations by doing all that we can today to change behavior.

Five Steps to Becoming a Cool Congregation

The graph on this page was created using data provided by the Environmental Change Institute, School of Geography and the Environment, and Department of Physics, University of Oxford, with support from the Oxford Martin Program on Resource Stewardship and the Union of Concerned Scientists. *Modeling by Richard Millar. Interactive by Duncan Clark.*



Temperature projections are based on the idealized climate model of Boucher and Reddy (2009), as used for calculation of warming potentials in IPCC (2013), driven by the IPCC "RCP8.5" high emissions scenario. They are consistent with, but not identical to, to the projections of the IPCC (2013).

Déjà vu?

Have we been here before? You may remember that once before we had a global scale crisis. Scientists told us that the ozone layer was disappearing, and we were told that life on Earth could not exist without the ozone's sun protection. The United States, and then the world, finally banned and stopped using all chemical substances that were harming the ozone layer. It took many years, and much hard work, but when everyone did their part, the ozone layer began to repair itself. Scientists now predict that by mid-century, the ozone hole will be a thing of the past. Success! Can we do it again? That is the hope and that's the challenge.

Nothing will motivate you, your team, or your volunteers more in becoming a Cool Congregation, than understanding what's at stake. You can make a real difference in what happens in the future, and perhaps the biggest impact will be as a role model and inspiration to others. It's the multiplier effect we're going for! So thank you for doing your part. In the Handouts section, you will find the latest Interfaith Power & Light Climate Change Facts. This also contains many useful links to reports and online interactive tools that aid us in understanding the changes we are now experiencing. Use this information to educate your team about climate change.

Form a Team

They go by many names

Speaking of a team, you're going to need one! They go by many names: green team, stewardship council, environmental task force, sustainability group, eco-group, or Creation care team are just a few of the names used. Your congregation may already have one, in which case, you will need to bring this idea to them. If not, you can form your own team. Here are some tips for forming and sustaining an enduring green team for the purposes of becoming a Cool Congregation.

Your "team" is perhaps the most important thing

The big picture of what we are trying to accomplish is gradually changing the culture of our faith communities so that caring for Creation becomes a central part of what it means to be a person of faith. We aim to bind care of Creation to the heart of our congregational life together. Cool Congregations offers us a hands-on way to accomplish this. You can't bring

about this cultural shift alone, who would want to? Many hands make light work — and besides, it's just more fun that way.

Determine how your team fits into the congregation's organizational structure

This will vary for each faith community. Larger congregations may have a committee for social concerns and form a subcommittee to be the Green Team. Smaller congregations may choose to have their Green Team be an independent committee. Many have reported the best success by receiving formal approval from the governing body of their congregation and by collaborating with other committees. Cool Congregations is a stewardship program that seeks to expand the concept of stewardship to include caring for Creation. It's meant to add a new dimension to your congregation's regular financial stewardship program, so be sure to include a member of the stewardship committee from your congregation on your Green Team.

Establish good communications with leaders in your congregation

Be sure to offer regular updates on your team's activity to your faith leaders, the governing board, and other committees by reporting in person or placing written updates in your congregation's newsletter, bulletins and website. Take the time to get your Green Team established as a regular part of your congregation's life. It will pay off later in wider involvement by the community in your Cool Congregations program. You may even want to ask your faith leader to "commission" your team as a part of a worship service.

Make sure your team is well-rounded

In addition to having someone on the team with good communications skills, you will need someone with basic math and computer skills (for helping to calculate energy savings and carbon footprint reductions), perhaps someone who is good with finances, another that knows a bit about raising money, and a people person who can rally some volunteers to your cause.

Some teams that form around Cool Congregations have received funding from their congregation to carry out their activities as these teams work on saving energy at the congregational facility and reducing the congregation's energy bills. Others raise funds in creative ways. These are discussed in Step 3 of this guide.

Finding members for your team

Here are a few ideas that others have found useful:

- ◆ Ask your faith leader to join or recommend names, including a lay governing board member.
- ◆ Show a movie on climate change or care for Creation to your congregation. Following the film, give a short talk about your Cool Congregations team and invite others at the screening to join you.
- ◆ Make an oral announcement during your worship service inviting interested people to attend a short meeting to learn about becoming a member of the team and about plans to become a Cool Congregation.
- ◆ Post a flyer, or put a similar announcement in your congregation's weekly bulletin or e-mail.
- ◆ Have each of your present team members invite one other person to your next meeting. If you shape your team with care, it will be strong and will endure.

Think About Your Goals

After you have found or formed your team, sit down together to reflect on why your congregation would like to become a Cool Congregation. Thinking about your goals will allow you to be able to write a mission statement. See sample mission statements on following pages.

Caring for Creation and the climate as acts of faithful stewardship is a relatively new idea. Some say it is a remembering of a very old idea. You will likely have to persuade others that this is an important part of living out your faith day-by-day. To do so, you need to be able to talk about what and why you're doing what you are. To be an effective agent of social change, your team won't need to be experts in climate change science or its solutions, but you will need a genuine, heartfelt story. Here are a few ideas to help you hone your team's mission statement:

Reflect on Personal Motivations

What are your personal motivations?

Write down your own reasons for wanting to help your faith community become a Cool Congregation. Share your thoughts with others on the team, and get their reaction and thoughts.

- ◆ What concerns you about climate change? What have you seen or heard about how climate change is affecting people you love or are concerned about — your grandchildren, your children, the poor or marginalized in your state and around the world?
- ◆ What is it you love about this beautiful Earth, our common home? Tell about a shaping experience in nature that you've had, a place on earth you treasure, or a place you go to renew yourself in nature. We are more likely to protect something that we love.
- ◆ What is your personal spiritual motivation to take action? Tell about how your religious experience or beliefs have motivated you to act.

Effectively engage your congregations by creating solid connections between your faith tradition and the Earth stewardship activities of the work of the Cool Congregations program. Crafting and communicating a narrative that is both authentic and empowering is an important first step.

Align with Denomination's Climate Statement

Is your denomination active on climate? Many religions have developed a stance on climate change or Creation care. Understanding what mission statements, programs, and resources already exist within your own denomination will be a big leg up.

Looking at denominational statements on caring for Creation and climate change are helpful in demonstrating a congregation is responding to a call that their faith tradition has already made. A stewardship mission is not a new thing at all; they have been part of faith traditions for a very long time.

Visit www.coolcongregations.org for links to Religious Statements on Climate Change under Resources. Members of your team can divide up the study on:

- ◆ Your denominational statement on caring for Creation and climate change.
- ◆ Your denominational leaders' statements on caring for Creation and climate change.
- ◆ Your congregation's faith leaders' views on the topic.
- ◆ Readings from other writers both inside and outside of your tradition.

Write Your Mission Statement

Mission statements are succinct statements that help guide goal setting and action plans. They will often include or refer to the denomination's statement on climate or care of Creation. Be sure to pass your green team's mission statement by the powers that be in your congregations for feedback and to get the buy-in you need to move forward with your projects. The goal of green teams is to green every decision made at congregation.

Sample Green Team or Congregational Mission Statements

- ◆ Recognizing that the 202nd General Assembly of the Presbyterian Church declared "Restoring Creation as a central concern of the church to be incorporated into its life and mission at every level," the Environmental Stewardship Committee commits:
 - To inspire members to respond to the Bible's call to appreciate, respect and responsibly care for God's gifts of His Earth and all His Living Creations;
 - To provide members with information and resources to preserve God's Creation;
 - To network with neighboring congregations and community organizations to promote stewardship of God's Creation.
- ◆ We will promote care for the earth and its resources as God's gifts for sustaining life. We will incorporate care for God's Creation throughout the St. Andrew's community, including our worship services, our educational programs, our outreach and service to our community, and our buildings and grounds.

Five Steps to Becoming a Cool Congregation

- ◆ Brith Shalom, as both a physical place and spiritual community, seeks to create a space of Torah, chesed, and kavannah committed to Jewish values. One of these values is tikkun olam, restoring a world damaged by human actions. By infusing environmental considerations and sustainability in all of our endeavors, we can take steps to repair the Creation that we celebrate. We also keep in mind the mitzvah of Bal Tashchit – not wasting the limited resources we have been given — as we think of ways to reduce, reuse, and recycle. Each act and decision, no matter how small, moves us in the direction of becoming a sustainable community.

As tempting as it is to skip this step because it's a lot of reflection - **it is essential to articulate the faith-based reasons for action to convince your congregation to make the necessary upfront financial commitment to fund the improvements to the facility.**

Step 2

Find Ideas

For your first project, choose one with a quick payback like changing lighting over to LED to demonstrate the concept of upfront investment with a payback through cost savings on the energy bill. A congregation that understands payback is more likely to invest in larger changes. Every project is an opportunity to educate your congregation. Be public about what you are doing and why!

Go electric! And source your electricity from renewable resources such as geothermal, air source heat pumps, solar thermal, solar electric, and wind. As utilities add more renewables to their mix and solar prices come down, it's more feasible than ever to go carbon free by going electric.





Interfaith Power & Light

COOL CONGREGATIONS

Project Ideas & Stories for Inspiration

Cool Congregations find unique expression from place to place. We've collected a few sample stories of what others have done to inspire you. Find more at CoolCongregations.org.

Cool Congregations Planners

North West Unitarian Universalist Congregation in Atlanta, Georgia submitted a comprehensive plan that included: environmental Sunday services; a stewardship project involving the planting of 10 threatened Eastern Hemlock saplings; a rainwater capture project; installation of motion sensors on the lights; and, solar panel array expansion.

The senior youth group of **United Parish of Upton, Massachusetts** embarked on an audit and planning project to save energy and money for their church. Working with National Grid, they raised more than \$5,000 for the recommended improvements. They also enacted skits before worship services to promote the call to have congregants follow suit at home through MASS SAVE.

Energy Savers

Mercy Center in Madison, Connecticut has curbed energy use and reduced six tons of carbon dioxide from entering the atmosphere through solar-powered hot water heaters that save 650 gallons of heating oil, programmable thermostats, attic and wall insulation, low-flow toilets, and lighting upgrades.

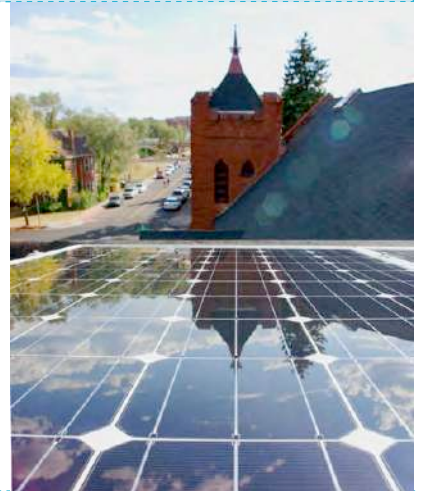
Jonesville United Methodist Church in Clifton Park, New York has done extensive work to lower its carbon footprint in response to climate change, such as: upgrading the sanctuary ceiling; replacing heating and cooling systems; and adding motion sensor and time controlled lighting. In all, Jonesville United Methodist Church is saving an estimated 18,000 kWh of electricity, 2,200 therms of natural gas and more than \$5,000 in annual cost savings. This represents a saving of 62,000 pounds of CO₂.

Cost effective insulation improvements at **Congregation Beth El in Bangor, Maine** have cut the energy purchased to power the congregation in half and at the same time made their building more comfortable, useful and more welcoming to their community.

Renewable Role Models

Federated Community Church in Flagstaff, Arizona meets in a century-old building that now gets 96% of its power from the sun. Congregants can monitor the system's performance via a link on the church website.

Euclid Avenue United Methodist Church in Oak Park, Illinois installed geothermal for heating and cooling. They have reduced their fossil fuel emissions by 100% and have reduced their energy costs by nearly \$12,000 per year. They host workshops in conjunction with Illinois Interfaith Power & Light at Faith in Place to inspire other churches and synagogues to implement geothermal.



Sacred Grounds Stewards

Good Samaritan Church in Pinellas Park, Florida developed a system for using recycled water for outdoor gardens. They also started using a natural blend of things like soap and Listerine as insecticides for the new native plants in their gardens.

University United Methodist Church in Austin, Texas is an urban church with a number of social justice ministries, including Open Door, a Saturday brunch for about 300 people experiencing homelessness. By adding commercial composting services and training its homeless clients, Open Door now runs trash free. It has composted more than 27,000 gallons of waste. The church has also quadrupled its recycling program as it works to serve vulnerable neighbors in the most healthy efficient, and environmentally sound means possible.

Community Inspiration

An energy study from **First Universalist Church of Rockland, Maine** showed significant heat loss through windows. The church saved 25% in its heating costs in the first winter. The project has grown into WindowDressers, which have installed energy saving window inserts in more than 520 homes in 42 Maine communities.

First Parish Unitarian Universalist of Medfield, Massachusetts conducted a community anti-idling campaign. They formed a coalition of community organizations, police, schools to garner media and get permanent signs erected by the police to prevent tons of carbon emissions.

Trinity Church, an Episcopal Community in Menlo Park, California hosted an "Energy Party," which focused primarily on educating members and inspiring them to go green at home.

Go Electric! Choose 100% renewable energy and switch all appliances to electric.

25 STEPS UNDER \$25

For Congregations

Lighting

CO2 Reduction

- ☐ 1. Turn off the lights in unused rooms.
- ☐ 2. Buy energy-efficient LED bulbs for the lights you use most. They last 10x longer and have no mercury.
- ☐ 3. Replace your halogen lamp with an LED lamp. Saves on air conditioning as LEDs are 90% cooler than halogen.
- ☐ 4. Replace your holiday decorations with LED lights.

380 pounds a year
193 pounds a year per bulb
475 pounds a year
122 pounds a season

Appliances & Gadgets

CO2 Reduction

- ☐ 5. Unplug and recycle old or rarely used refrigerators.
- ☐ 6. Allow laundry items to air-dry after washing.
- ☐ 7. Cut your phantom electric loads in half by installing surge protectors for computers, copier, fax machine, printer, TV, etc. and turning off your office equipment when you're not using it. Make sure sound systems and projectors are turned off when not in use.
- ☐ 8. Run dishwasher only with a full load and use the energy-saving setting to dry dishes. Or don't use heat when drying, just open the door to air dry.
- ☐ 9. Use electric lawn and garden care equipment. Even if your electricity comes from coal it's still less polluting.

2,500 pounds a year
200 pounds a year for every dryer load reduced per week;
780 pounds a year for the entire summer;
1,400 pounds a year if year-round
A minimum of 500 pounds a year. (Phantom loads account for 6% of our nation's electrical use.)
200 pounds a year
At least 100 pounds a year

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Learn more at coolcongregations.org



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Heating & Cooling	CO2 Reduction
☐ 10. Ask utility for a building energy audit to find out how to improve your building's efficiency. These audits are usually free and make you eligible for rebates on improvements.	Potentially thousands of pounds a year
☐ 11. Don't overheat or overcool rooms. Adjust thermostat lower in winter and higher in summer. Close curtains to block sunlight in summer.	500 pounds a year for each 2 degree adjustment
☐ 12. Clean or replace air filters as recommended. Cleaning a dirty air conditioner filter can save 5% of the energy used.	175 pounds a year
☐ 13. Install programmable thermostats to automatically adjust temperatures.	600 pounds a year
☐ 14. Caulk and weather-strip around doors and windows to plug air leaks.	Up to 1,350 pounds a year
Water	CO2 Reduction
☐ 15. If your congregation contains showers, install shower timers to encourage each shower user to reduce their shower time by 5 minutes.	250 pounds a year per person
☐ 16. Instead of washing laundry items in hot water, wash them in cold water.	500 pounds a year for each two loads a week reduced
☐ 17. Turn down water heater's thermostat to 120 degrees. Think about installing a programmable thermostat on water heater if you only need hot water on worship days.	500 pounds a year for each 10 degree adjustment
☐ 18. Install low-flow showerheads and faucets to use less hot water.	Up to 300 pounds a year
☐ 19. Install low-flow faucet aerators.	20 pounds a year per faucet
☐ 20. If water heater is 10 years old, or more, wrap it in an insulating jacket.	Up to 1,000 pounds a year
Getting Around	CO2 Reduction
☐ 22. Whenever possible, encourage congregants to walk, bike, carpool, or use mass transit. Develop a carpooling system.	20 pounds for every gallon of gas saved
☐ 23. Check the inflation in congregation vehicle tires to increase fuel efficiency.	250 pounds a year
☐ 24. Change congregation vehicle air filters according to the car manual.	200 pounds a year
Reduce Food Waste	CO2 Reduction
☐ 25. Food waste causes methane in the landfill that is 22x more potent than CO2. Plan congregational meals to avoid leftovers, buy only what you need, donate leftovers (Good Samaritan law allow this.)	100 pounds a year for every gallon per week reduced
Cool Congregations Challenge is a program of Interfaith Power & Light . More info at CoolCongregations.org	

Go Electric! Choose 100% renewable energy and switch all appliances to electric.

25 STEPS OVER \$25

For Congregations

Lighting

CO2 Reduction

- ☐ 1. Lighting typically accounts for 30% to 50% of energy use in most buildings. Replace standard incandescent bulbs with LED bulbs, which use 85% less energy and last up to 10 times longer.
- ☐ 2. Replace incandescent exit signs with LED exit signs.
- ☐ 3. Open curtains to use natural lighting in the sanctuary, install sun tubes and sky lights to bring in natural sunlight. Be sure to equip with remote control blinds.
- ☐ 4. Install occupant sensors where lights tend to get left on.

193 pounds per bulb per year

500 pounds per year per sign

Saves in direct proportion to electric-generated lighting it replaces

As much as 50% of lighting related CO₂ emissions

Appliances

CO2 Reduction

- ☐ 5. Replace older refrigerator, washer and dryer, or dishwasher with ENERGY STAR models.

2,300 pounds a year, 1,300 pounds per year, 120 pounds per year respectively

Heating & Cooling

CO2 Reduction

- ☐ 6. Upgrade air conditioning system with ENERGY STAR.
- ☐ 7. Add or improve the insulation of walls and ceilings. This can save about 25% on heating bills.
- ☐ 8. If you need a new furnace, install an air or ground source heat pump heating and cooling system.
- ☐ 9. If you need to replace your windows, install the best energy-saving models.

Min. of 185 pounds a year

Up to 2,000 pounds a year

7,000-14,000 pounds a year

Up to 10,000 pounds a year

COOL CONGREGATIONS

Learn more at coolcongregations.org



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- ☐ 10. Plant trees next to your building. Evergreens on the north and west to break cold winds, and deciduous trees on the south and west for cooling shade.
- ☐ 11. Add air-gap window films to seal leaky windows in winter and/or add low-e films to keep out summer heat.
- ☐ 12. Add insulation to your basement.
- ☐ 13. Seal and insulate warm air heating ducts.

About 2,000 pounds a year

400 and 380 pounds a year, respectively

750 pounds a year

800 pounds a year

Water

CO2 Reduction

- ☐ 14. Replace your standard electric hot water heater with an "on demand" hot water system.
- ☐ 15. Add on solar water heater to reduce use of gas hot water heater. Gas hot water heaters are responsible for 50% of gas usage in a home.
- ☐ 16. Add the hot water "desuperheater" to your geothermal system to pre-heat the water in your electric hot water heater.
- ☐ 17. Replace auto-flush toilets with dual flush toilets. Install motion-activated water faucets on sinks.
- ☐ 18. Reduce size of manicured lawns with native grasses and plants that require less water and maintenance, and store more carbon.

3,600 pounds a year

2300 pounds a year

Reduces water heating carbon by 55%

Cut toilet-related water use in half

Lawns use up to 20 times more water

Getting Around

CO2 Reduction

- ☐ 19. When ready, replace the congregation's car, bus, or van with a more efficient model.
- ☐ 20. Buy a hybrid vehicle. The average driver saves \$3,750 a year.
- ☐ 21. Switch to all electric vehicles and install charging stations.

About 10,000 pounds a year for each additional 10 mpg

16,000 pounds a year for the average driver

Charge vehicles with zero carbon renewable energy.

Renewable Energy

CO₂ Reduction

- ☐ 22. Install a solar thermal heating system for space heating. It's the most cost effective way to use solar for space heating.
- ☐ 23. Install solar electric system. Couple with air or ground source heat pump heating and cooling for a zero carbon building.
- ☐ 24. Install a wind turbine for electric generation
- ☐ 25. Plant and protect trees in the Global South to offset remaining carbon emissions through IPL's Carbon Covenant program. www.co2covenant.org

Potentially thousands of pounds for each of these actions.

Cool Congregations Challenge is a program of [Interfaith Power & Light](http://InterfaithPower.org). More info at CoolCongregations.org

Step 3 Get Help

When you are part of Interfaith Power & Light's Cool Congregations, there are always people to help. Use the resources below to find the help you need to plan and complete your project.

The Cool Congregations Website

There are many helpful resources at www.coolcongregations.org including a list of inspirational stories from other congregations that outline what they did, how they did it, and what impact their projects had on energy and carbon reduction, and cost savings.

IPL State Network

Interfaith Power & Light recommends that you contact the Interfaith Power & Light affiliate in your state for ideas and possible assistance. There are IPL affiliates in 40 states, and they can provide the best tips on where to find help for your project. Many of them offer very useful services, workshops, conferences, and referral lists. To locate your state affiliate, visit the website www.coolcongregations.org/ipl-in-your-state/



US Green Building Council

IPL has formed a partnership with the US Green Building Council (USGBC) through our state affiliate system. USGBC has successfully run the LEED (Leadership in Energy and Environmental Design) certification program for buildings, homes and communities since 2000. Through your IPL affiliate, you may be able to find USGBC volunteers or professionals to perform an energy audit or to help you develop an action plan. Certified Cool Congregations and winners of the Cool Congregations Challenge are entered in the USGBC's *Green Building Information Gateway* website. Inspiring other congregations with their success. www.gbiv.org/activities/

EPA Energy Star for Congregations

IPL has formed a partnership with the Environmental Protection Agency's Energy Star for Congregations program. The EPA has many resources for congregations, including Portfolio Manager, which is discussed in greater details in the Step 3 section of this guide. Find the immensely useful Energy Star for Congregations Action Workbook with an introduction from Interfaith Power & Light with language you can use to educate your congregation at

www.coolcongregations.org/resources/helpful-links

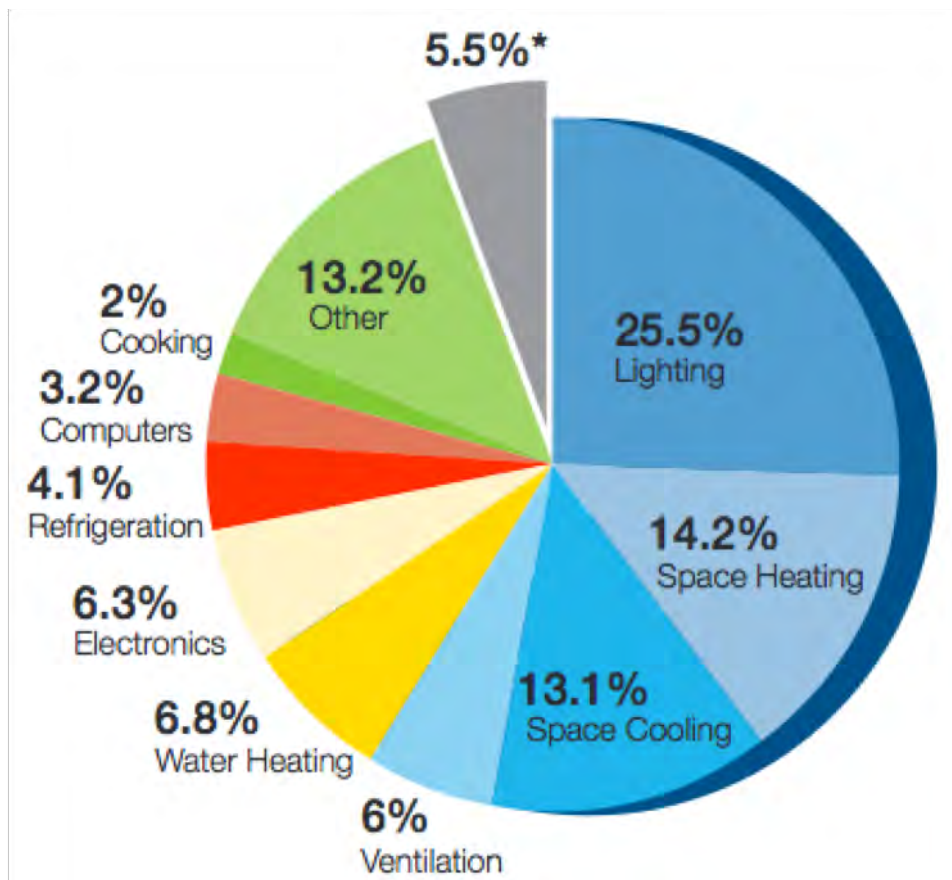
It is a perfect companion to the Cool Congregations program. Visit the Energy Star for Congregations website to find a myriad of other resources and tools for your next project. We have included their flyer in the following pages so you can learn more about the resources the EPA can provide to congregations.

www.energystar.gov/congregations

Step 4

Make Improvements

According to the Energy Information Administration pie chart below for commercial buildings, including houses of worship, the three largest energy demands are lighting, heating, and cooling. Water heating, ventilation, and non-PC office equipment are each around 6 percent of the total, followed by refrigeration, computer use, and cooking. If you want to make a big dent in your energy consumption, take on the largest pie sections first.



** This pie chart includes one quad of energy (5.5 percent) that is a statistical adjustment by the Energy Information Administration to reconcile two divergent data sources.*

What is a Carbon Footprint?

The vast majority of energy we use today is derived from the burning of fossil fuels such as oil, natural gas, or coal. A carbon footprint, as commonly known, is simply the weight of carbon or carbon dioxide emitted into the atmosphere each year from the burning of fossil fuels. Other gases that contribute to climate change — such as methane from waste — are converted into "carbon equivalent" units then added to the carbon footprint.

Determine Your Carbon Footprint with Our Calculator

Use our Cool Congregations Calculator, found at coolcongregations.org to estimate your congregation's carbon footprint. It offers a snapshot of your carbon footprint, allowing you to look at the best places to lower your footprint and begin to work towards zero carbon. We break your carbon footprint down into four sub-components:

Facilities:

Carbon emissions from your congregation's use of electricity, natural gas and other fuels, waste and water.

Transportation:

Carbon emissions from congregants travel to worship and staff travel.

Food:

Emissions from various types of foods served at congregational meals.

Goods and Services:

Carbon emissions associated with office products, cleaning products, and everything else your congregation purchases.

Then we give credit for any carbon offsets your congregation has purchased.

EPA Energy Star Portfolio Manager Tool

Another calculator that congregations find very useful is the **ENERGY STAR's Portfolio Manager** — the Environmental Protection Agency's free, in-depth online data management tool that helps facilities managers and/or homeowners track and measure a building's energy use. This program is not an Interfaith Power & Light program, however, many congregations use it, and we've partnered with Energy Star to pass along Portfolio Manager information and news.

Portfolio Manager is especially recommended for congregations with someone on staff or in a volunteer capacity who is web / tech-savvy. It can calculate the energy intensity of a particular building while taking into account changes in weather. Comparisons are made with other buildings of the same type, same building material and age to establish a national "average". Once a baseline is established with the data for your building, Portfolio Manager can perform various calculations and generate reports that "rate" your building, determine the percent energy use reduction per year, and track progress towards your goals. ENERGY STAR Portfolio Manager classifies major commercial building types, such as hospitals, schools, houses of worship or congregations, restaurants, industrial space, etc. They also hold an annual "Battle of the Buildings" for those that use the tool, and many IPL Cool Congregations have entered.

You will need to know the amount of energy your congregation consumed during three of the coldest or hottest months of the year in your area.

- ◆ Kilowatt Hours of Electricity
- ◆ Million BTUs of Natural Gas
- ◆ Gallons of Heating Oil
- ◆ Pounds of Coal

These can usually be retrieved from energy bills, or from your utility company's online website. Alternatively, you can estimate cost savings, and document this percentage, taking into account any major increase or decrease in the price of energy during the measurement period.

Set a Reduction Goal

The next step is to set an energy reduction goal. This goal should be realistic yet challenging. It should be specific in terms of desired energy savings and deadlines should be established. A clear goal will help rally your congregation's stakeholders and provide a measure against which you can evaluate your progress.

An energy reduction goal usually has several parts:

- ◆ A Baseline (e.g. current energy use before improvements)
- ◆ A Reduction Amount (e.g. reduce by 20%)

Interfaith Power & Light encourages congregations to set as a default goal something equal to or stronger than the EPA's ENERGY STAR National *Battle of the Buildings* Competition, which is to reduce energy use by 20%.

Create your action plan

An action plan specifically spells out how your team will achieve or begin to achieve your energy reduction goal. This is where specific energy conservation measures should be defined — like replacing old, incandescent light bulbs. The action plan should state who will be responsible for accomplishing each measure and what resources they will have available to support them. Use Cool Congregation's "25 Steps Under \$25" and "25 Steps Over \$25" checklists in Step 2 for ideas. Also, refer to Cool Congregation feature stories for inspiration. Find them at www.coolcongregations.org.

A good action plan includes:

- Mission or reasons for project
- Reduction goals (energy, costs, carbon)
- Itemization of reduction strategies (replace sanctuary lighting, etc)

If you need help putting together your action plan, try reaching out to your state chapter of Interfaith Power & Light, which has affiliates across the US. Visit our website www.coolcongregations.org/ipl-in-your-state/ to learn about the local resources and programs to assist you on your path to becoming a Cool Congregation. In the “Find Help” section of this guide, we list other organizations that may be able to provide assistance.

Implement your action plan

The implementation process inevitably involves unexpected surprises and demands some flexibility. Be sure to keep energy management team members informed of progress and setbacks. Communicate frequently with all congregation stakeholders about your energy management plan so they understand the bigger picture and how it relates to changes they see occurring.

Measure and report your energy saving performance

The only way your congregation will know if your program achieved success will be through a consistent and regular process of measuring and reporting. By regularly reporting your energy use, you will actually encourage progress. Demonstrating early success leads to higher engagement and desire to participate among stakeholders. Your measurement reports should be easily understandable to all stakeholders, especially those with little or no background in energy efficiency.

Record and save your “Baseline Carbon Footprint”

Knowing your baseline will allow you to calculate your total carbon footprint reduction, or percent (%) reduced. This will be important when continuing on to Step 5, applying for Cool Congregations certification.



The WindowDressers program was started at The First Universalist Church in Rockland, Maine — a Cool Congregation. As of 2013, WindowDresser have installed energy saving window inserts in more than 520 homes in 42 Maine communities, saving customers a projected \$4,400 (each) over the next decade and reducing nearly 4,600 metric tons of CO2 emissions by 2020.

Five Steps to Becoming a Cool Congregation

Step 5

Apply for Certification

Becoming a Certified Cool Congregation is a wonderful way for faith communities to showcase their achievements and model positive solutions to the challenges of climate change.

A Certified Cool Congregation is a house of worship of any religion that has successfully cut its carbon footprint by 10% or more in response to the threat of climate change. In addition to receiving a Certified Cool Congregations paper certificate for framing, you will become a national role model with a listing on the coolcongregations.org website, inclusion in the US Green Building Council's Green Building Information Gateway, and have the option of purchasing an attractive, eco-friendly bamboo wall plaque to showcase your achievements to all who enter your congregation.

Your congregation's energy savings can come from any combination of the following: energy efficiency, energy conservation, or use of renewable energy. See Step 4 to find ways to measure your "before" and "after" snapshot of your congregation's energy use and carbon emissions to determine your reductions for the application for certification

The application for Cool Congregations certification is online. Review the instructions below, to make your application process flow easily. Find the application online:

<https://www.coolcongregations.org/get-certified/>

LEVELS OF CERTIFICATION

Work your way up from a 10% cut in carbon reductions, all the way up to 40%, 60%, 80% or even 100%! The application will ask you to specify what level of certification you seek. You can apply for 10% this year, and 20% next, or you might be ready to apply for the 40% reduction certification now. Here are the levels:

- 10% or more – Red
- 20% or more – Blue
- 30% or more – Green
- 40% or more – Bronze
- 60% or more – Silver
- 80% or more – Gold
- 100% or more – Platinum

THE APPLICATION

Certification applications are accepted on a rolling basis. Your project can be a recently completed project or a one completed in prior years. You must be able to provide “before” and “after” energy use figures so we can help determine the level of certification.

Please review the application questions below and answer them (and keep a record of your answers) before you open the online application. You must complete the application in one sitting. You will not be able to save a draft and come back to it later.

The online application is found here: <https://www.coolcongregations.org/get-certified/>

Once your congregation is certified, IPL will post a story about your accomplishment to the Certified Congregations webpage and you can link to it from your congregation’s website to share your success!

There is no fee to enter -- although we do request that participating congregations consider making a [donation to Interfaith Power & Light](#) from your energy cost savings to “pay it forward” so that we can continue to help more congregations go green.

All fields are required. Applications are accepted on a rolling basis. Please allow 6 weeks for your application to be reviewed. We will notify you directly of your certification status as soon as it becomes available. For questions contact programs@interfaithpowerandlight.org

Application questions are below.

1. Do you have permission to apply on behalf of your congregation?
2. Full name of congregation as you would like it to appear on the certificate.
3. Average number of members of your congregation.
4. Religion or denomination
5. Address of congregation
 - a. Street address
 - b. Street address Line 2
 - c. City
 - d. State
 - e. Zip

6. Setting
 - a. Urban
 - b. Rural
 - c. Suburban
7. Project Start Date (estimates ok)
8. Project End Date (estimates ok)
9. What committee or team worked on this project? Please tell us a bit about them, and mention any names or credits here.
10. Please describe your energy saving project (ie. What challenge were you faced with and how did you meet it? Final results are asked in a narrative question below) (250 word limit) ex. Challenge: an aging boiler, concern about climate change, high energy bills, lightning struck our rooftop AC unit, etc. Met the challenge by deciding to install geothermal, or go all-electric powered by solar, etc.
11. Cost of your project (estimates ok)
12. Annual cost savings of your project (estimates ok)
13. Baseline usage of electricity from the grid/utility, expressed in kWh per year. Provide an annual figure. (Baseline year refers to BEFORE improvements)
14. Grid/Utility Electrical usage AFTER improvements, expressed in kWh per year. Provide an annual figure.
15. Baseline fuel usage, expressed in therms (gas, oil, coal). Provide an annual figure. (Baseline year refers to BEFORE improvements)
16. Fuel usage AFTER improvements, expressed in therms. Provide an annual figure.
17. Please enter surplus electricity from your own solar or wind in kWh/year generated for which you are being credited. If not applicable, list NA.
18. If you purchase "green" electricity from a company that sells electricity sourced from renewables through a special contract with your congregation, (must be 5 years in length or longer to count for this certification) please provide average annual usage in kWh. Please attach proof of contract to this application. If not applicable, list NA.

19. If you purchase electricity through a solar contract such as a power purchase agreement or a community solar agreement (must be 5 years in length or longer to count for this certification) please provide average annual usage in kWh. Please attach proof of contract to this application. If not applicable, enter NA.
20. If you purchase carbon offsets (must be a commitment of 5 years in length or longer to count for this certification) please provide average annual figure in kWh avoided. If not applicable enter NA. Please attach proof of purchase to this application.
21. How many tons of carbon emissions per year has your congregation reduced on average compared to before the project began? Please measure the combined carbon footprint of electricity and gas to figure the difference. Report the savings in tons. Use the Cool Congregations Congregational calculator to determine before and after figures for total CO2 reduced. coolcongregations.org
22. Carbon calculator used:
- a. Cool Congregations Calculator
 - b. EPA's Energy Portfolio
 - c. Other
 - d. None, did calculations by hand
23. What percent reduction or certification level would you like to be considered for by Interfaith Power & Light?
- a. 10%
 - b. 20%
 - c. 30%
 - d. 40%
 - e. 60%
 - f. 80%
 - g. 100%
24. What motivated your congregation to undertake this project? (ex. a sermon, a study group on climate, high energy bills, concern for the climate your grandchildren will inherit, concern for the poor who suffer the worst from climate impacts?....) (250 word limit)
25. Describe the final results of your project. (Help us understand how you measured the success of your project, and its impact on the climate. Tell the story about cost and carbon and energy savings, who was involved in terms of volunteers, a green team, the community, what you did, and so forth.) (250 wd limit)
26. What will others find inspiring about your project? (What about this project will others find particularly inspiring?) (250 word limit)

27. Has your congregation ever participated in any of these programs that you know of?

- Faith Climate Action Week
- Cool Congregations Challenge
- Carbon Covenant
- None of the above

28. Has your congregation participated in climate advocacy or environmental justice programs or sermons?

- Yes
- No
- We're thinking about it

29. Attach photos and accompanying documents that you would like to provide. Photos will be used for Cool Congregations.

30. Add your web address if desired.

The online application is found here: <https://www.coolcongregations.org/get-certified/>

COOL CONGREGATIONS CERTIFICATION POLICY

Permanent changes to the facility

IPL counts 100% of the reductions in energy use due to permanent changes made to the congregational facility and its operations towards certification. IPL places the greatest emphasis on making permanent changes to the facility to save energy through energy efficiency or energy conservation, or through installing renewable energy sources.

Solar contracts

IPL also acknowledges congregations who engage in long-term (at least 5 years) contracts for renewable energy through power purchase agreements or community solar contracts. 100% of these reductions count towards a congregation's certification level. Please attach proof of contract to this application. (It could be a copy of the contract or an energy bill with the name of the provider.)

Green electricity contracts

IPL acknowledges and certifies purchases of 100% renewable electricity from a utility company in long-term contracts (at least 5 years) by adding up to 5% to the certification level. Renewable electricity is electricity supplied through the grid in a special contract with a utility provider that provides 100% electricity from renewable sources, such as East Bay Community Energy (ebce.org) For example: if a congregation has reduced electricity use by 50%, and has purchased 100% renewable electricity for the remaining 50% of electricity use, IPL will award up to an additional 5% certification level for a total of 55% certification. Please attach proof of contract to this application. (It could be a copy of the contract or an energy bill with the name of the provider.)

Carbon offsets

IPL acknowledges the purchase of carbon offsets and will add up to 5% to a congregation's total certification level from the purchase (for a commitment of at least 5 years.) For example: if a congregation has reduced by 50%, and has purchased carbon offsets for the remaining 50%, IPL will award up to an additional 5% certification level for the offsets for a total of 55% certification. Please attach proof of purchase to this application.

As of May, 2021

Thank you for your interest in becoming a Certified Cool Congregation!

Handouts

*Go the extra mile.
Lead the way to a brighter future.
Win \$1,000*

COOL CONGREGATIONS CHALLENGE

The **Cool Congregations Challenge** is a contest hosted by Interfaith Power & Light that awards prizes to congregations that are becoming energy efficient and sustainable role models in response to climate change. Projects completed by December 15 qualify. Apply at www.coolcongregations.org/take-the-challenge/

\$1,000 prize will go to the winning contestant in each category:

**Cool Congregations Planner
Energy Saver
Renewable Role Model
Sacred Grounds
Community Inspiration**



Interfaith Power & Light
A Religious Response to Global Warming

Cool Congregations Challenge is a program of Interfaith Power & Light.
More info at coolcongregations.org.



LEARN MORE AT
energystar.gov

ENERGY STAR® is a U.S. Environmental Protection Agency program helping businesses and individuals fight global warming through superior energy efficiency.

STEWARDSHIP OF THE EARTH: STEWARDSHIP OF YOUR FUNDS

Virtually all faith traditions teach stewardship of financial and, especially, of natural resources which support life on Earth. Energy efficiency improves financial stewardship by reducing utility costs. Greater efficiency in your worship facility, members' homes and workplaces can also help protect the Earth for future generations and make life better for everyone today.

PUTTING ENERGY INTO STEWARDSHIP

The U.S. Environmental Protection Agency's mission protecting human life and health is likely a good match for your stewardship concerns. EPA's **ENERGY STAR for Congregations** provides free information, technical support, and public recognition for congregations that take significant action against energy waste. ENERGY STAR estimates that if America's 370,000 congregations cut energy use just 20% it would save nearly \$630 million for missions and other priorities, while providing more than 3.6 billion "new" kWh electricity without additional cost or pollution. These energy savings would prevent more than 2.6million tons of greenhouse gas emissions – the equivalent of eliminating emissions from 480,000 cars, or planting 60,000 trees.

A Call to Action

ENERGY STAR is ready to work with your congregation. Your congregation can:

- Become an ENERGY STAR Partner at www.energystar.gov/join
- Benchmark and track your energy performance, money saved and pollution prevented with a free, online tool at www.energystar.gov/benchmark
- Find free information and learn what others have achieved at www.energystar.gov/congregations
- Learn about home energy efficiency at www.energystar.gov/homeimprovement





LEARN MORE AT
energystar.gov

ENERGY STAR® is a U.S. Environmental Protection Agency program helping businesses and individuals fight global warming through superior energy efficiency.

Take Action in Your Worship to Save Energy—Some First Steps...

OFFICE AND KITCHEN EQUIPMENT

Always buy ENERGY STAR qualified equipment for your congregation, business and home. The ENERGY STAR label indicates the most efficient computers, printers, copiers, televisions, windows, ceiling fans, and other appliances and equipment.

LIGHTING

Turn off lights (and other equipment) when not in use. High utility costs often include paying for energy that is completely wasted.

Install “occupancy sensors” to automatically turn off lighting when no one is present, and back on as people return.

Replace incandescent bulbs with ENERGY STAR qualified compact fluorescent lamps (CFLs) and/or light emitting diodes (LEDs) CFLs cost about 75% less to operate, and last about 10 times longer; LEDs last about 20 times longer.

HEATING AND AIR CONDITIONING

Keep exterior doors closed while running your HVAC: It's simple, but not always done and it will help to avoid wasteful loss of heated or cooled air.

Change HVAC filters (or clean if reusable) every month during peak cooling or heating season. Dirty filters waste energy, overwork the equipment, and cause poor indoor air quality. New filters are cheap.

Install a quality programmable thermostat to optimize the operation of your HVAC system on a “24/7” basis for savings, comfort and convenience.

Use fans. Comfort is a function of temperature, humidity, and air movement. Fans can help delay or reduce the need for air conditioning by making a higher temperature feel more comfortable.

Plug leaks with weather-stripping and caulking for controlled ventilation to stop outside air infiltration. Many air leaks and drafts are easy to find because they are easy to feel—like those around windows and doors - but holes hidden in HVAC ducts, attics, basements, and crawlspaces can be bigger problems.

Set up a yearly “pre-season” maintenance contract for “tune-up” of your HVAC system before heating and cooling seasons to protect performance.

WATER - HOT AND COLD

Fix leaks. Small leaks add up to many gallons and dollars wasted - especially for hot water.

Buy an ENERGY STAR qualified water heater when a new one is needed. Consider a “tankless” water heater to reduce “standby” storage costs and waste.

Set water temperature at about 110-120 degrees to prevent dangerous scalds and save energy.

Learn more at www.energystar.gov/congregations



EPA's 1 – 100 ENERGY STAR Score and Your Facility's Energy and Financial Performance

To get a clear idea of how efficiently your facility uses energy, you can “benchmark” in EPA's free, online Portfolio Manager tool. From this baseline, you can plan, manage, and track improvement projects toward success. ENERGY STAR partners know that “you can't manage what you don't measure.”

Portfolio Manager helps you assess whole building energy performance; track changes in energy, water use and costs and greenhouse gas emissions over time; track green power purchases; create custom reports; share data with others and see how your building compares to worship facilities nationwide. With this information, your congregation can make more informed decisions on energy-efficient investments and keep everyone well informed about your progress stewarding financial and natural resources.

Why should I use Portfolio Manager?

The tool calculates and updates a simple, easy to interpret 1 - 100 ENERGY STAR score based on your utility data, and key facility factors such as square footage, equipment and hours of operation. These factors and weather are “normalized” for changes over time, with your score accurately documenting savings. Scores of 75 or higher are eligible for national ENERGY STAR certification. For more information on Portfolio Manager and to create an account, please see www.energystar.gov/benchmark.